

Marianne Dehasque, PhD

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Ecology and Evolutionary Biology
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Employment history:

2025 - present	Postdoctoral Researcher Supervisor: Malin Pinsky UC Santa Cruz, CA, USA
2023 - 2025	Grant holder (Carl Tryggers Stipendium) Supervisor: Torsten Günther Uppsala University, Sweden
2018- 2023	PhD student Stockholm University, Sweden
2022- 2023	Lab manager Centre for Palaeogenetics, Sweden
2017-2018	Jr. Data Architect Agilent technologies, Multiplicom branch, Belgium

EDUCATION:

2018 - 2023	PhD in Systematic Zoology: Defended March 10, 2023 Title: <i>Palaeogenomic reconstruction of woolly mammoth evolutionary history and extinction dynamics</i> Supervisor: Love Dalén Stockholm University, Sweden
2011-2016	BSc & MSc in Biology University of Antwerp, Belgium

PEER-REVIEWED PUBLICATIONS (*indicates shared first authorship)

- Mármol-Sánchez, E., Fromm, B., Oskolkov, N., Pochon, Z., **Dehasque, M.**, Aslanzadeh, M., Bozlak, E., Brown, K., van der Valk, T., Kalogeropoulos, P., Chacón-Duque, J.C., Biryukova, I., Heintzman, P.D., Furugård, C., Plotnikov, V., Protopopov, A., Andersson, B., Ersmark, E., Peterson, K.J., Friedländer, M.R., Dalén, L., 2025. Ancient RNA expression profiles from the extinct woolly mammoth. *Cell* 0. <https://doi.org/10.1016/j.cell.2025.10.025>
- Dehasque, M.**, van der Valk, T., Chacón-Duque, J.C., Termes, L., Larsson, P., Moots, H.M., Tubbesing, F., Larsdotter, J., Oteo-García, G., Moreland, K., van Essen, H., Arbour, V., Keddie, G., Richards, M.P., Díez-Del-Molino, D., Heintzman, P.D., Lister, A., Dalén, L., 2025. Genomic and morphological analysis reveals long-term mammoth hybridization in British Columbia, Canada. *Biol. Lett.* 21, 20250305.

- Guinet, B., Oskolkov, N., Moreland, K., Dehasque, M., Chacón-Duque, J.C., Angerbjörn, A., Arsuaga, J.L., Danilov, G., Kanellidou, F., Kitchener, A.C., Muller, H., Plotnikov, V., Protopopov, A., Tikhonov, A., Termes, L., Zazula, G., Mortensen, P., Grigorieva, L., Richards, M., Shapiro, B., Lister, A.M., Vartanyan, S., Díez-Del-Molino, D., Götherström, A., Pečnerová, P., Nikolskiy, P., Dalén, L., van der Valk, T., 2025. Ancient host-associated microbes obtained from mammoth remains. *Cell* 188, 6606–6619.e24.
- Chacón-Duque, J. C., Thomas Thorpe, J. A., Li, W., **Dehasque, M.**, Pečnerová, P., Barlow, A., Díez-Del-Molino, D., Henneberger, K., Jin, C., Moreland, K. N., Paijmans, J. L. A., van der Valk, T., Westbury, M. V., Wijnands, F., Barnes, I., Germonpré, M., Hall, E., Hewitson, S., Mol, D., ... Dalén, L. (2025). A million years of mammoth mitogenome evolution. *Molecular Biology and Evolution*, 42(4), 1–9.
- Atağ, G., Kaptan, D., Yüncü, E., Başak Vural, K., Mereu, P., Pirastru, M., ... **Dehasque M.**,... Somel, M. (2024). Population Genomic History of the Endangered Anatolian and Cyprian Mouflons in Relation to Worldwide Wild, Feral, and Domestic Sheep Lineages. *Genome Biology and Evolution*, 16(5), evae090.
- **Dehasque, M.**, Morales, H.E., Díez-del-Molino, D., Pečnerová, P., Chacón-Duque, J.C., Kanellidou, F., Muller, H., Plotnikov, V., Protopopov, Tikhonov, A., A., Nikolskiy, P., Danilov, G.K., Gianni, M., van der Sluis, L., Higham, T., Heintzman, P.D., Oskolkov, N., Gilbert, M.T.P., Götherström, A., van der Valk, T., Vartanyan, S , Dalén, L., 2024. Temporal dynamics of woolly mammoth genome erosion prior to extinction. *Cell*, 187(14), 3531-3540.
- Sandoval-Velasco, M., Dudchenko, O., Rodríguez, J. A., Estrada, C. P., **Dehasque, M.**, Fontseré, C., ... & Aiden, E. L. (2024). Three-dimensional genome architecture persists in a 52,000-year-old woolly mammoth skin sample. *Cell*, 187(14), 3541-3562.
- Díez-del-Molino, D., **Dehasque, M.**, Chacón-Duque, J.C., Pečnerová, P., Tikhonov, A., Protopopov, A., Plotnikov, V., Kanellidou, F., Nikolskiy, P., Mortensen, P., Danilov, G.K., Vartanyan, S., Gilbert, M.T.P., Lister, A.M., Heintzman, P.D., van der Valk, T., Dalén, L., 2023. Genomics of adaptive evolution in the woolly mammoth. *Current Biology*, 33(9), pp.1753-1764.
- Kutschera, V.E., Kierczak, M., van der Valk, T., von Seth, J., Dussex, N., Lord, E., **Dehasque, M.**, Stanton, D.W.G., Khoonsari, P.E., Nystedt, B., Dalén, L., Díez-del-Molino, D., 2022. GenErode: a bioinformatics pipeline to investigate genome erosion in endangered and extinct species. *BMC Bioinformatics* 23, 1–17.
- van der Valk, T. *, **Dehasque, M.***, Chacón-Duque, J.C., Oskolkov, N., Vartanyan, S., Heintzman, P.D., Pečnerová, P., Díez-Del-Molino, D., Dalén, L., 2022. Evolutionary consequences of genomic deletions and insertions in the woolly mammoth genome. *iScience* 25, 104826.
- **Dehasque, M.**, Pečnerová, P., Lagerholm, V.K., Ersmark, E., Danilov, G.K., Mortensen, P., Vartanyan, S., Dalén, L., 2022. Development and Optimization of a Silica Column-Based Extraction Protocol for Ancient DNA. *Genes*. 13(4), 687.
- Dussex, N., Bergfeldt, N., de Anca Prado, V., **Dehasque, M.**, Díez-Del-Molino, D., Ersmark, E., Kanellidou, F., Larsson, P., Lemež, Š., Lord, E., Mármol-Sánchez, E., Meleg, I.N., Måsviken, J., Naidoo, T., Studerus, J., Vicente, M., von Seth, J., Götherström, A., Dalén, L., Heintzman, P.D., 2021. Integrating multi-taxon palaeogenomes and sedimentary ancient DNA to study past ecosystem dynamics. *Proc. Biol. Sci.* 288, 20211252.
- Rosengren, E., Acatrinei, A., Cruceru, N., **Dehasque, M.**, Haliuc, A., Lord, E., Mircea, C.I., Rusu, I., Mármol-Sánchez, E., Kelemen, B.S., Meleg, I.N., 2021. Ancient Faunal History Revealed by Interdisciplinary Biomolecular Approaches. *Diversity*. 13(1), 1-17.
- Pečnerová, P., Díez-Del-Molino, D., van der Valk, T., **Dehasque, M.**, Götherström, A., Dalén, L., 2021. *Mammuthus* sp. (Early and Middle Pleistocene Mammoths). *Trends Genet.* 37, 682–683.

- **Dehasque, M.**, Pečnerová, P., Muller, H., Tikhonov, A., Nikolskiy, P., Tsigankova, V.I., Danilov, G.K., Díez-del-Molino, D., Vartanyan, S., Dalén, L., Lister, A.M., 2021. Combining Bayesian age models and genetics to investigate population dynamics and extinction of the last mammoths in northern Siberia. *Quaternary Science Reviews*. 259, 106913.
- van der Valk, T., Pečnerová, P., Díez-Del-Molino, D., Bergström, A., Oppenheimer, J., Hartmann, S., Xenikoudakis, G., Thomas, J.A., **Dehasque, M.**, Sağlıcan, E., Fidan, F.R., Barnes, I., Liu, S., Somel, M., Heintzman, P.D., Nikolskiy, P., Shapiro, B., Skoglund, P., Hofreiter, M., Lister, A.M., Götherström, A., Dalén, L., 2021. Million-year-old DNA sheds light on the genomic history of mammoths. *Nature* 591, 265–269.
- Feuerborn, T.R., Palkopoulou, E., van der Valk, T., von Seth, J., Munters, A.R., Pečnerová, P., **Dehasque, M.**, Ureña, I., Ersmark, E., Lagerholm, V.K., Krzewińska, M., Rodríguez-Varela, R., Götherström, A., Dalén, L., Díez-Del-Molino, D., 2020. Competitive mapping allows for the identification and exclusion of human DNA contamination in ancient faunal genomic datasets. *BMC Genomics* 21, 844.
- **Dehasque, M.**, Ávila-Arcos, M.C., Díez-Del-Molino, D., Fumagalli, M., Guschanski, K., Lorenzen, E.D., Malaspinas, A.-S., Marques-Bonet, T., Martin, M.D., Murray, G.G.R., Papadopoulos, A.S.T., Therikildsen, N.O., Wegmann, D., Dalén, L., Foote, A.D., 2020. Inference of natural selection from ancient DNA. *Evol Lett* 4, 94–108.

SCHOLARSHIPS AND AWARDS

2025	Marcus Borgströms <i>Research grant, 200,000 SEK</i>
2023	Lars Hiertas Minne <i>Research grant, 21,000 SEK</i>
2023	Nilsson-Ehle stipendium <i>Research grant, 93,000 SEK</i>
2023	SMBE Graduate Student Excellence Award, SMBE23, Ferrara, Italy <i>The SMBE Graduate Student Excellence Award honors the best presentation at the Graduate Student Excellence symposium.</i>
2023	Liljevalchs resestipendier <i>Travel stipendium, 7,000 SEK</i>
2023	Albert & Maria Bergströms Stiftelse <i>Travel stipendium, 10,000 SEK</i>
2022	First prize for best student talk at ConsGen22, Scotland
2020	Appointed a personal advisor in the Bioinformatics Advisory Program, Sweden. This is a competitive call.
2020	Riksmusei Vänner Stipendium, Sweden <i>Travel stipendium, 10,000 SEK</i>
2019, 2020	Alice and Lars Silén Fond, Sweden <i>Travel stipendium, 5,500 and 8,000 SEK resp.</i>

2016 Finalist at KETS awards, 2016, Belgium
Scientific award for best Belgian thesis in zoology

SELECTED PRESENTATIONS

2025 Ecology and Evolutionary Biology Seminar, UC Santa Cruz, USA
Invited speaker: *Woolly mammoth evolution and extinction through the lens of ancient DNA*

2025 Evolution in Sweden, Linköping, Sweden
Oral presentation: *From wild to tamed and back – The evolutionary history of Sardinian sheep*

2024 AaRC TikTalks
Invited speaker: *Temporal dynamics of woolly mammoth genome erosion prior to extinction*

2023 SMBE Satellite Meeting on Molecular Evolution in Small populations, Princeton University, USA
Invited speaker: *Temporal dynamics of woolly mammoth genome erosion prior to extinction*

2023 PAG 30, San Diego, USA
Oral presentation: *Temporal dynamics of woolly mammoth genome erosion prior to extinction*

2022 ConsGen22, Edinburgh, Scotland
Oral presentation: *The extinction of the woolly mammoth. What can we learn?*

2021 ISBA 9, Virtual conference
Oral presentation: *Studying selection in mammoths using genome-wide time series data.*

2021 Polar forum, Stockholm, Sweden
Invited oral presentation: *Mammoth Models and How to Date them.*

2020 Inaugural symposium of the Centre for Palaeogenetics, Stockholm, Sweden
Invited oral presentation: *Evolution and extinction of the woolly mammoth.*

2019 ESEB, Turku, Finland
Poster presentation: *Optimization of double-stranded library preparation methods for ancient and degraded DNA.*

TEACHING

2025 Training on Demographic inference from genetic data, Cebu City, Philippines
Course organiser and lecturer

2023 Palaeontology master's course, Swedish Museum of Natural History
Lecturer: *Permafrost Palaeogenomics*

2022 Methods in molecular life science, Stockholm University

Lecturer: *Permafrost Palaeogenomics*

2019-2021 Conservation Genomics, Stockholm University
Lecturer: *Genetics and extinction, Ancient DNA in conservation*

2018-2020 Palaeoecology, Genetics, and Human Prehistory, Stockholm University
Lecturer: *Using ancient DNA to study selection*
Assistant wet-lab part

SUPERVISION

PhD students

2025-present Mikaela Salvador, PhD mentorship
2023-2024 Viviana Florida, PhD mentorship

Master and Bachelor students

2020-2021 Foteini Kanellidou, MSc thesis supervision
2018 Héloïse Müller, BSc internship supervision

PROFESSIONAL SERVICE AND OUTREACH

Journal refereeing: Molecular Ecology Resources, Current Biology, BMC Ecology and Evolution, BMC Biology, Cell, Nature Communications, Scientific Reports, Genome Biology and Evolution

2025 Popular science talk on mammoths for Creswell Crags
(<https://www.creswell-crags.org.uk/>)

2024 Participation in YouTube video "*The Strange DNA of the last mammoths*".
(<https://www.youtube.com/watch?v=mf1HWjUJPZM>)

2024 Wrote a popular science article for The Conversation
Did inbreeding cause the woolly mammoth's extinction? Our research suggests it was more sudden than that

2024 Organizer of symposium at SMBE24, Puerto Vallarta, Mexico
Decoding the past to safeguard the future

2022 Popular science lecture on mammoths for classroom of 16-year-olds.

2019 Participation in Discovery Channel documentary "*Lost Beasts of the Ice Age*". Aired 2019.

2019 Organized Gästabudet
Gästabudet is the annual symposium for PhD students from the Zoology department.

2018,2019 Organized two LET'S (Life Earth Time 'n Space) seminars.
LET's is a seminar series hosted by the Swedish Museum of Natural History to a broader audience interested in life sciences.